Transformers and Attention-Based Deep Networks Assignment - 3

Due: 17.05.2026

May 5, 2026

Introduction

Large Vision-Language Models (VLMs), such as PaliGemma, which combines SigLIP and Gemma, have shown strong performance in image captioning tasks. However, fine-tuning these models for domain-specific applications remains computationally expensive due to their scale.

Parameter-Efficient Fine-Tuning (PEFT) methods address this limitation by adapting models with minimal trainable parameters. In particular, QLoRA enables efficient fine-tuning through low-rank adaptation combined with quantization, significantly reducing memory requirements.

In this work, we apply QLoRA to fine-tune PaliGemma on a remote sensing image captioning dataset, aiming to efficiently adapt the model to domain-specific visual and semantic characteristics.

Assignment

Dataset

You will be using the RISC dataset Figure 1, uploaded for you to caglarmert/full_riscm. There are 44521 remote sensing images (satellite) with fixed 224x224 resolution and 222605 captions (5 per image) briefly summarizing the content of the image.

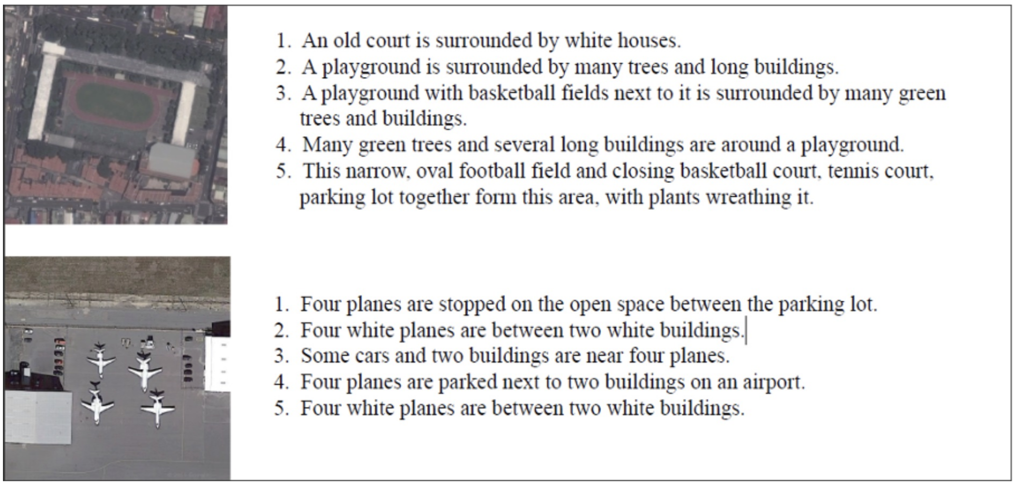

Figure 1: Remote sensing image captioning dataset

Vision-Language Model

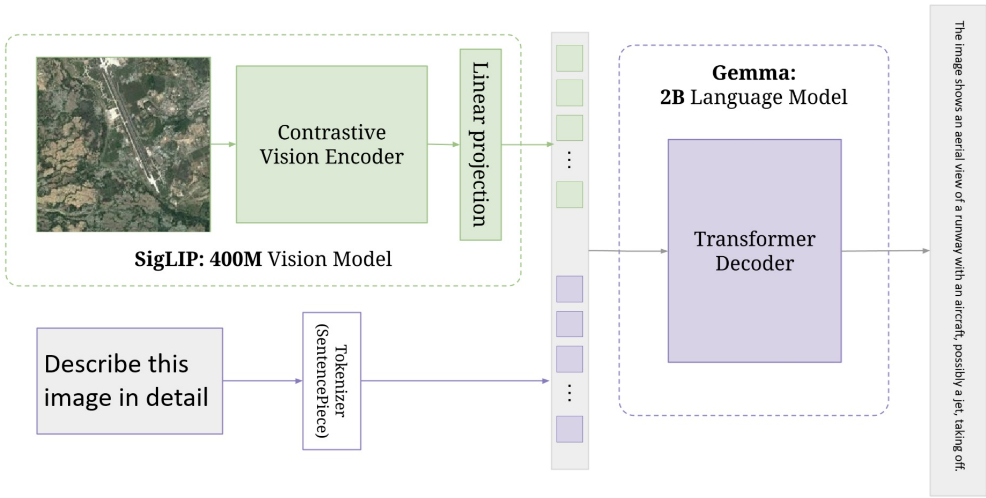

Figure 2: PaliGemma Architecture

In this project, we adopt **PaliGemma**, a vision-language model combining **SigLIP** (vision encoder) and **Gemma** (language decoder), as illustrated in Figure 2, as the baseline architecture. The main objective is to fine-tune (efficiently) this model on the image captioning task.

It is important to note that PaliGemma inherits limitations from both its vision encoder (SigLIP) and language decoder (Gemma). Common limitations of Vision-Language Models (VLMs) include:

- Sensitivity to prompt design; clear and structured instructions are often required.
- Limited capability in handling highly creative or complex reasoning tasks.
- Difficulty in understanding nuanced language, such as sarcasm or figurative expressions.
- Potential to generate hallucinated or factually incorrect outputs, as they are not grounded knowledge bases.
- Reliance on statistical correlations rather than true common-sense reasoning.

PaliGemma is:

- A general-purpose pre-trained model suitable for downstream fine-tuning tasks.
- A strong baseline for adapting vision-language models to domain-specific applications.

PaliGemma is not:

- A zero-shot optimal solution; specialized models may outperform it without fine-tuning.
- A conversational chatbot designed for multi-turn dialogue.

Deliverables

Your submission is expected to be uploaded to ODTUCLASS in a single zip file containing your report and code.
Report: Prepare a short report (1-page IEEE format) briefly explaining your steps and demonstrating results.
Code: The reproducible, clean and working code in .ipynb format with outputs saved.

Late submission after the due date is available for a penalty of 10 points for each day.

Grading

1. **Parameter Efficient Fine Tuning (40 pts)**: Apply parameter efficient fine tuning (QLoRA) to the pre-trained Pali-Gemma 3B model, as shown in the lab-5, use the full dataset.
2. **Results (30 pts)**: Demonstrate at least 5 example images, with one of the ground truth captions of your choice, baseline PaliGemma result, PEFT applied PaliGemma result.
3. **Performance Metric (30 pts)**: Find a suitable performance metric for caption evaluation. Select one of the ground-truth captions (out of 5 available) as your target and compare the baseline PaliGemma and QLoRA applied PaliGemma results.